KAUST

AMCS206 Applied Numerical Methods

Problem Set 3

March 7, 2026

Problem 1

Find the solution to the Poisson equation

$$-\frac{d^2}{dx^2}u(x) = x^2(1-x)$$

in the unit interval $[0,1]$ using homogeneous Dirichlet conditions on the two boundaries, $u(0) = u(1) = 0$. Use a uniform a grid of size $n = 100$. Solve the system using (i) Jacobi, and (ii) Gauss-Seidel iterations. Plot (semilogy) the residual as a function of iteration number. Is this behavior expected and why?

Problem 2

Let $t_1, \ldots, t_n$ be n numbers. The Vandermonde matrix is defined as

$$A = \begin{bmatrix} 1 & t_1 & \cdots & t_1^{n-2} & t_1^{n-1} \\ 1 & t_2 & \cdots & t_2^{n-2} & t_2^{n-1} \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & t_n & \cdots & t_n^{n-2} & t_n^{n-1} \end{bmatrix}$$

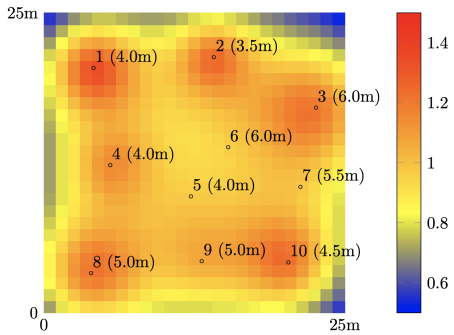
i.e., $A_{ij} = t_i^{j-1}$. Construct A for the vector $t = 1 : .5 : 8.5$ and estimate its condition number using `condest`. Let $\mathbf{1}$ be a vector whose entries are all 1. Compute a vector $b = A * \mathbf{1}$ and use it to compute the solution of $Ax = b$. Compute the relative residual error $||Ax - b||/||b||$. Does this imply that the relative error in the solution is accurate? Explain briefly why or why not?

Problem 3

A set of n lamps illuminates an area that we divide into m regions or pixels. We let l_i denote the lighting level in region i so the vector $l_{m\times 1}$ gives the illumination levels across all regions. We let p_i denote the power at which lamp i operates, so the vector $p_{n\times 1}$ gives the set of lamp powers.

The vector of illumination levels is a linear function of the lamp powers, so we have $l = Ap$ for some $m \times n$ matrix A. The jth column of A gives the illumination pattern for lamp j, i.e., the illumination when lamp j has power 1 and all other lamps are off. We will assume that A has linearly independent columns (and therefore is tall). The ith row of A gives the sensitivity of pixel i to the n lamp powers. The illumination decays with an inverse square law, so A_{ij} is proportional to d_{ij}^{-2}, where d_{ij} is the (3-D) distance between the center of the pixel and the lamp position.

Your task is to find lamp powers that result in a illumination patters that is the closest to a uniform illumination pattern with value 1 in all pixels. You will find on Blackboard data for the example below (prob33.m) The area is 25×25 grid with $m = 625$ pixels, each 1m square and the lamps are at various heights ranging from 3m to 6m, and at the positions shown.



Problem 4

In least squares, the objective (to be minimized) is

$$||Ax - b||^2 = \sum_{i=1}^{m} (a_i^T x - b_i)^2,$$

where a_i^T are the rows of A and the vector x is to be computed. In the *weighted least squares* problem, we minimize the objective

$$\sum_{i=1}^{m} w_i (a_i^T x - b_i)^2,$$

where w_i are given positive weights. This allow us to assign different weights to the different components of the residual vector.

1. Show that the weighted least squares objective can be expressed as $||D(Ax - b)||^2$ for an appropriate diagonal matrix D. This allows us to solve the weighted least squares problem as a standard least squares problem, by minimizing $||Bx - d||^2$, where $B = DA$ and $d = Db$.

2. The least squares solution is given by $\tilde{x} = (A^T A)^{-1} A^T b$. Give a similar formula for the solution of the weighted least squares problem. You might want to use the matrix $W = \mathbf{diag}(w)$ in your formula.

Problem 5

We would like to fit a polynomial with p terms (i.e., of degree $p-1$) to a set of m points. The polynomial is of the form

$$\hat{f}(x) = \sum_{i=1}^{p} \theta_i f_i(x)$$

where the basis functions are chosen as the monomials $f_i(x) = x^{i-1}$.

1. Fit a degree 2 polynomial to the $m = 100$ point data set defined in the file prob35.m on Blackboard. Plot the polynomial and compute the RMS error of the fit.

2. Repeat the fit above with degree 6 and degree 15 polynomials.

3. Which one of the three polynomials above would you recommend to use as a reasonable representation of the underlying model that this data set came from?

Problem 6

A computer takes around one second to fit a regression model (using least squares) with 20 parameters using 10^6 data points.

1. About how long do you guess it will take the same computer to fit the same 20-parameter model using 10^7 data points (i.e., $10\times$ more data points)?

2. About how long do you guess it will take the same computer to fit a 200-parameter model using 10^6 data points (i.e., $10\times$ more model parameters)?

Problem 7

You are given data $x^{(i)} \in \mathbb{R}$ and $y^{(i)}$, $i = 1, \cdots, N$ (see file on Blackboard) that are believed to be originating from a sinusoid signal of the form:

$$\hat{f}(x) = a_1 \cos(\omega_1 x) + b_1 \sin(\omega_1 x) + a_2 \cos(\omega_2 x) + b_2 \sin(\omega_2 x)$$

with $\omega_1 = 2\pi$, and $\omega_2 = 4\pi$. Formulate and solve a least squares problem to minimize the RMS value of the approximation error r with components $r_i = y^{(i)} - \hat{f}(x^{(i)})$.

Problem 8

You are given data set consisting of 100 points (file prob38.m on Blackboard). We would like to try two different models for fitting the data.

1. Fit a linear function to the data set and compute the RMS value of the fit.
2. Add the two piecewise linear basis functions $(x-1)_+$ and $(x+1)_+$ (now you have 4 basis functions in total) and redo the fit. Compute the RMS value obtained.
3. Is the RMS of the second fit smaller than that of the first? Is this expected? Explain briefly.

Problem 9

You are given data $x^{(i)} \in \mathbb{R}^2$ and corresponding $z^{(i)}$, $i = 1, \cdots, N$ (file prob39.m on Blackboard). Generate a least squares fit of the data in the form

$$\hat{f}(x) = a_1 x_1 + a_2 x_2 + b$$

and compute the resulting RMS value of the approximation error.

Problem 10

The idea of polynomial models can be extended from the case where x^i are scalars. In this exercise we consider a quadratic (degree two) model with $x \in \mathbb{R}^3$. This has the form

$$\hat{f}(x) = a + b_1 x_1 + b_2 x_2 + b_3 x_3 + c_1 x_1^2 + c_2 x_2^2 + c_3 x_3^2 + c_4 x_1 x_2 + c_5 x_1 x_3 + c_6 x_2 x_3$$

where the scalar a, 3×1 vector b, and 6×1 vector c are the zeroth, first, and second order coefficients in the model. Put this model into our general linear form, by giving p, and the basis functions $f_1, \cdots, f_p$.